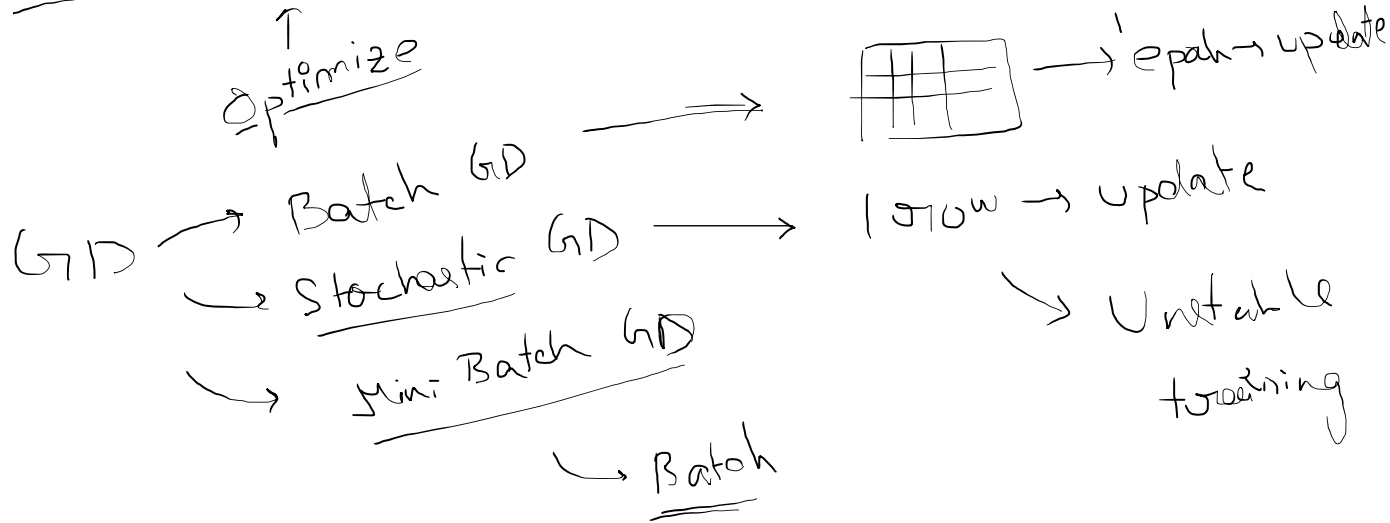
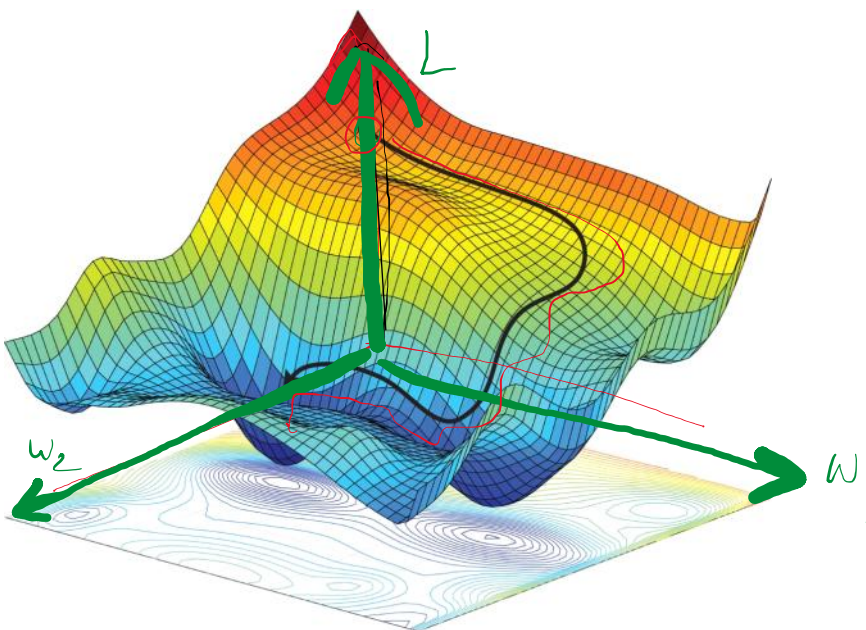


How are we implementing Gradient Descent ?



$$W = W - \eta \frac{dL}{dW}$$

Fast / slow we want move



$$L = f(w_1, w_2)$$

η → How fast / slow we want to move

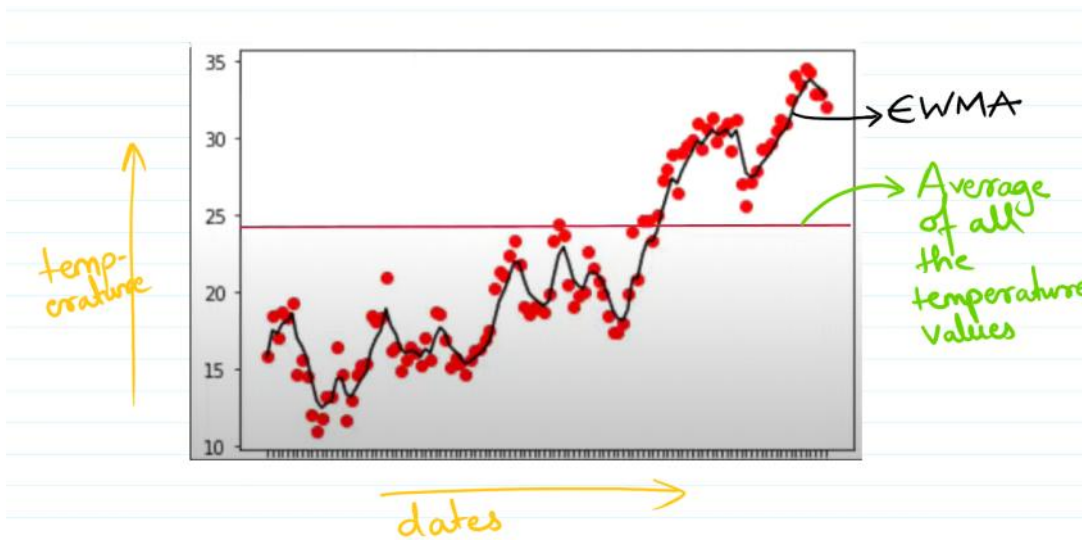
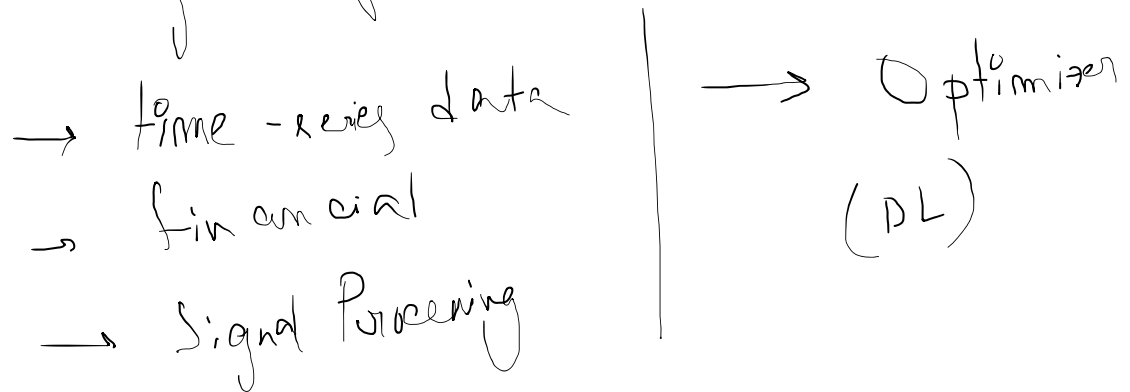


X Learning Rate Scheduling:-

Pre defined $\rightarrow$ [Small $\eta \rightarrow$ increase η]
 $\nwarrow$ Not data dependent

EWMA ✓
Momentum
NAG
RMSProp
Adam ✓
 $\nwarrow$ used

Exponentially Weighted Moving Average



$$V_t = \beta V_{t-1} + (1-\beta) \theta_t$$

data Attribute

EWMA at time t

$$0 < \beta < 1$$

$$0.9$$

$$\beta = 0.9$$

EWMA at $t=0$

$$V_0 = 0 \quad | \quad V_0 = \theta_0$$

$$V_0 = 25$$

$$\frac{\text{index}}{D_1}$$

$$D_2$$

$$\frac{\text{temp}(\theta)}{25}$$

$$13$$

$$V_1 = 0.9 * 25 + \frac{D_3}{(1-0.9) \times 25} \quad 1.7$$

$$V_1 = 1.3$$

$$V_2 = 0.9 * 1.3 + (1-0.9) * 13 \Rightarrow V_2 = 0.9 * V_1 + 1.3$$

$$V_3 = 0.9 * V_2 + (1-0.9) * 17$$

$$V_3 = 0.9 V_2 + 1.7$$

$$D_4 > D_3 > D_2 > D_1$$

$$\beta = V_t = \leftarrow \text{last}$$

$$\frac{1}{1-\beta}$$

intuition

$$\beta = 0.9$$

$$\frac{1}{1-0.9} = \frac{1}{0.1} = 10 \leftarrow \text{last 10 days average}$$

$$\beta = 0.98$$

$$\frac{1}{1-0.98} = \frac{1}{0.02} = 50 \text{ days "}$$

$$\beta = 0.5$$

$$\frac{1}{1-0.5} = \frac{1}{0.5} = 2 \text{ days "}$$

$$D_4 > D_3 > D_2 > D_1 ?$$

$$V_t = \beta V_{t-1} + (1-\beta)\theta_t$$

$$V_0 = 0$$

$$V_1 = (1-\beta)\theta_1$$

$$V_2 = \beta[(1-\beta)\theta_1] + (1-\beta)\theta_2$$

$$V_3 = \beta V_2 + (1-\beta)\theta_3$$

$$= \beta[\beta(1-\beta)\theta_1 + (1-\beta)\theta_2] + (1-\beta)\theta_3$$

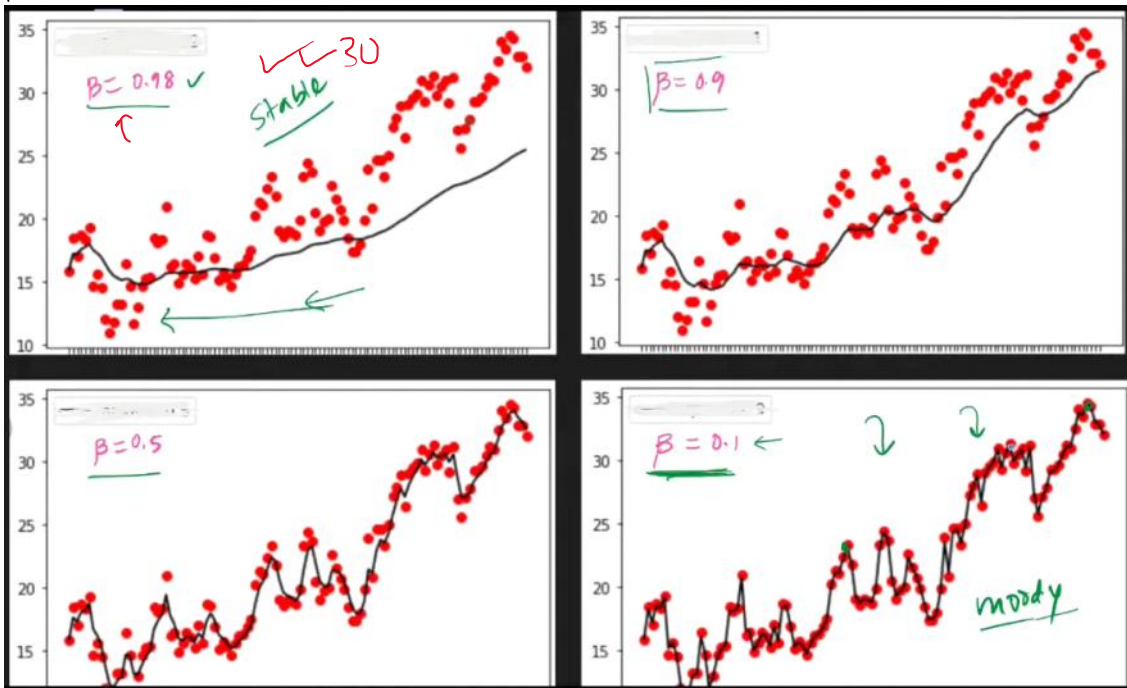
$$V_3 = \beta^2(1-\beta)\theta_1 + \beta(1-\beta)\theta_2 + (1-\beta)\theta_3$$

$$V_4 = \beta^3(1-\beta)\theta_1 + \beta^2(1-\beta)\theta_2 + \beta(1-\beta)\theta_3 + (1-\beta)\theta_4$$

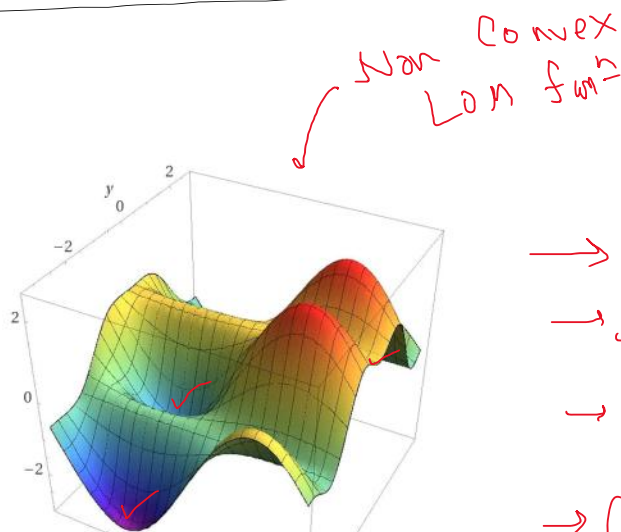
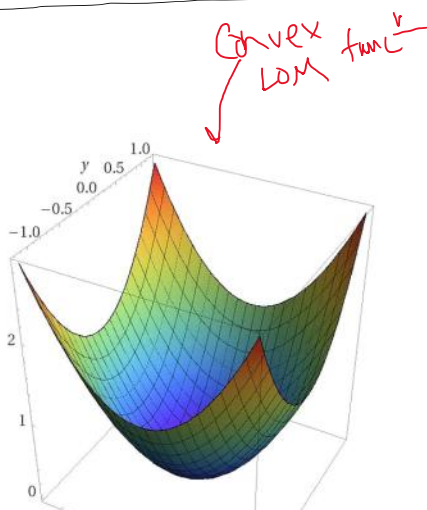
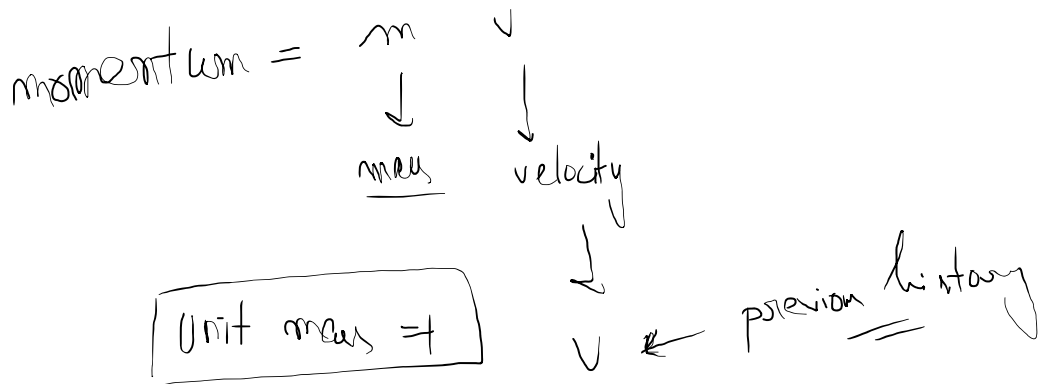
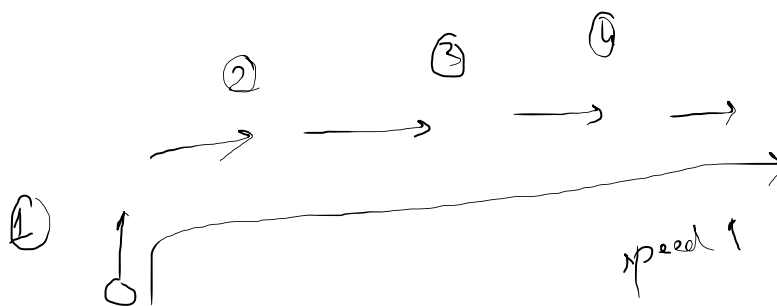
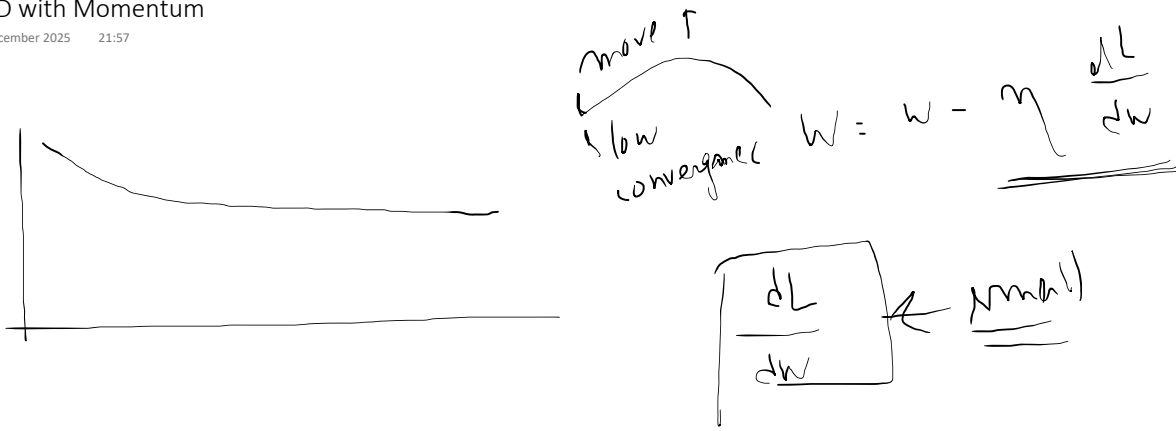
$$V_n = \frac{(1-\beta)}{\beta} \left[\frac{\beta^3\theta_1}{\beta} + \frac{\beta^2\theta_2}{\beta} + \frac{\beta\theta_3}{\beta} + \frac{\theta_4}{\beta} \right]$$

$$0 < \beta < 1$$

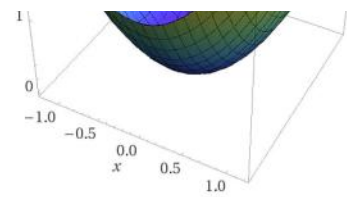
$$V_t = (1-\beta) \left[\beta^{t-1}\theta_1 + \beta^{t-2}\theta_2 + \dots + \beta\theta_{t-1} + \theta_t \right]$$



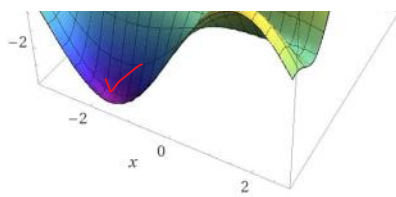
$$\frac{1}{|-0.1|} = \frac{1}{0.9} = \frac{10}{9} = \boxed{1.1}$$



- Local Minima
- Saddle point
- High Curvature
- Consistent

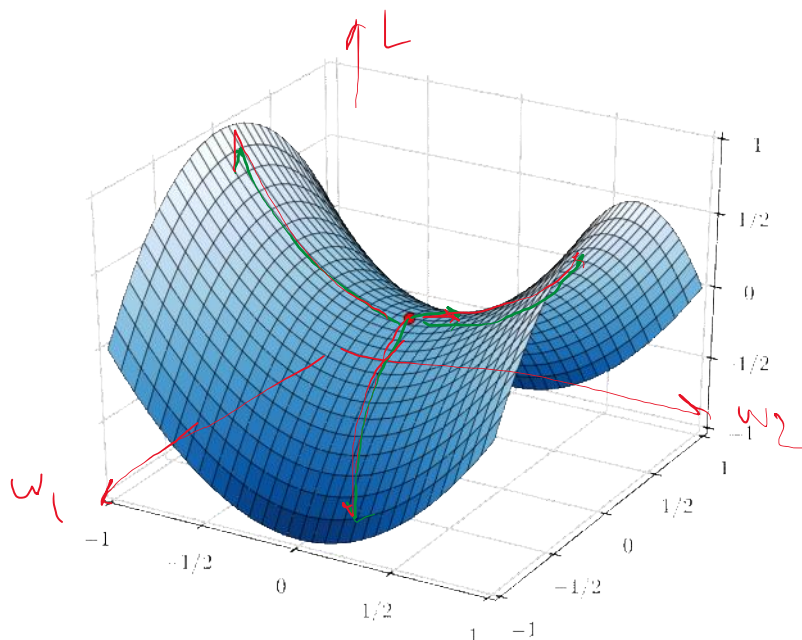
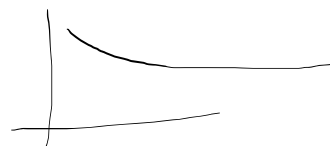


Computed by WolframAlpha

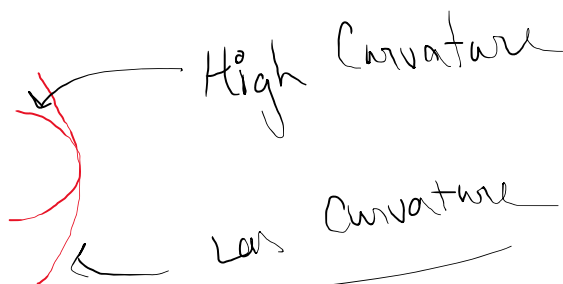


Computed by WolframAlpha

$\rightarrow$ Constant gradient



Saddle point



SGD with momentum $\gg$ SGD (99.9%)

$$w_{t+1} = w_t - \eta \Delta w_t \leftarrow \text{SGD}$$

EWMA at time t

EWIHT ~

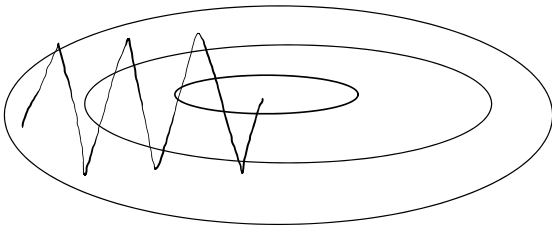
$$\Rightarrow \boxed{w_{t+1} = w_t - v_t}$$

$$v_t = \beta * v_{t-1} + \eta \Delta w_t$$

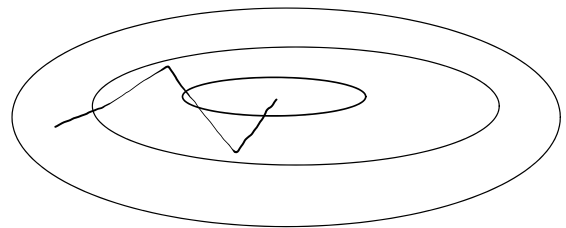
$$\boxed{w_{t+1} = w_t - (\beta v_{t-1} + \eta \Delta w_t)}$$

← SGD
with momentum

$$\beta=0 \Rightarrow w_{t+1} = w_t - \eta \Delta w_t \quad (\text{SGD})$$



(without momentum)



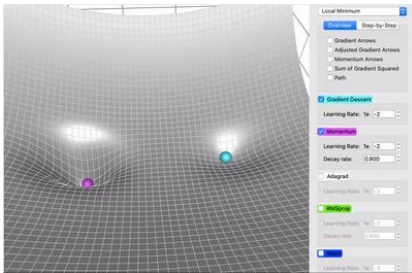
Higher Horizontal movement
(with momentum)

$\beta \leftarrow$ decaying factor

$$\beta_1 = 1 \quad \boxed{w_{t+1} = w_t - \nabla_{t-1} - \eta \Delta w_t}$$

↑ dynamic equilibrium / no fading

$$\boxed{0 < \beta < 1} \quad \beta = 0.9$$



<https://contributor.insightmediagroup.io/wp-content/uploads/2025/03/image-21.gif>

advantage
momentum
→ fast convergence

disadvantage
momentum
→ unnecessary oscillations